

Poppin Music — Design System

Variation D: Pure Engineering + Soft Skeuomorphism

Version 0 · September 2026

0. Philosophy

Variation D is Variation A (Pure Engineering) with a 40% skeuomorphic layer applied. The interface behaves like precision hardware — every surface has a physical orientation relative to an implied light source sitting above and slightly left. Depth cues are deliberate and minimal: a card feels like a piece of card stock on a warm cream desk, a button feels like a physical key, an input field feels pressed into the surface. Nothing is theatrical. A user might not consciously notice the depth; they will feel the difference.

1. Logo

Mark

A 2×2 pixel grid. Four equal squares arranged in a 2×2 layout with a 2px gap between them.

Position	Treatment
Top-left	Filled solid — #273038
Top-right	Filled solid — #273038
Bottom-left	Filled solid — #273038
Bottom-right	Outline only — 1px border #273038 , transparent interior

Sizes

- App header mark: 12×12px (each square ~5px, gap 2px)
- Favicon: 16×16px
- Standalone asset: 128×128px

Logo Philosophy

The mark is a 2×2 bit cell — the most fundamental unit of digital data. Three squares are filled; one is not.

The three filled squares are the Session Agents — Harmony, Bass, Drums. They are complete, defined, filled with technique and craft. The system provides.

The outlined square is the human. It has a boundary — intention, taste, the ear that decides — but its interior is always open. It is never filled in advance because the human is the creative authority, the one who listens and commits. The system cannot fill that square for you.

Together the four squares form a complete 2×2 byte — the irreducible unit of the product. The music is only whole when all four positions are occupied: three agents and one listener.

The gap between the squares (2px) is the silence between musicians. Space that is felt, not heard.

Usage Rules

- Always render on a transparent or single-colour background. No drop shadows on the mark itself.
- Minimum size: 12×12px (3-pixel squares with 1px gap) — do not render smaller or the outline square loses legibility.
- Colour: always #273038 on light backgrounds. On dark backgrounds invert to #E8E4DC.
- Never use the mark in a colour other than the primary text colour of its context.

2. Colour System

Backgrounds

Token	Hex	Usage
bg-page	#F5F2ED + radial gradient	Page/body background. Apply <code>radial-gradient(ellipse at 50% 0%, #F8F5F0 0%, #F0EDE6 100%)</code> — imperceptibly lighter at top-centre, implies an overhead light source.
bg-panel	#FBFAF7 + linear gradient	All cards, panels, forms, rails. Apply <code>linear-gradient(160deg, #FFFFFF 0%, #F7F5F0 100%)</code> .
bg-input	#EDEAE4	Text inputs, textareas. Receded into the surface.
bg-header	#F5F2ED	Top navigation bar. Same as page.

Text

Token	Hex	Usage
text-primary	#273038	Headings, active labels, all primary copy
text-secondary	#646A6B	Supporting body copy, descriptions
text-tertiary	#747A7D	Metadata, timestamps, contextual notes
text-muted	#8D908F	Inactive labels, placeholder-level text
text-faint	#999A96	Captions, version tags, idle states
text-inverse	#FBFAF7	Text on dark (#273038) button backgrounds

Borders & Dividers

Token	Hex	Usage
border-default	#D8D4CE	Standard panel borders, separators, rail dividers
border-strong	#BEBB5	Slightly stronger — input borders, candidate card borders, bottom command-bar border
border-active	#273038	Active/selected state (active section pill, currently-playing candidate card, focused input)

Semantic Agent Colours

Each Session Agent has one colour. The colour appears as: small square status icons (7px), left-edge agent card accent (3px border-left), MIDI note block fill, and tag backgrounds.

Agent	Base Fill	Gradient (MIDI blocks)	Usage
Harmony	#B8C8D4	linear-gradient(180deg, #BFCEDA 0%, #AEBECB 100%)	Dusty blue
Bass	#D4C8B8	linear-gradient(180deg, #DACFC0 0%, #C9BCAA 100%)	Warm sand
Drums	#D4B8C3	linear-gradient(180deg, #DABFCA 0%, #C9AAB6 100%)	Dusty rose
Committed / Keep	#B8D4C3	linear-gradient(180deg, #C3DDCD 0%, #ACCAB7 100%)	Sage green — used on “Keep this” button
Reaper Bridge	#C3B8D4	linear-gradient(180deg, #CABFDA 0%, #B5AAC9 100%)	Muted lavender — used on bridge status indicator

All agent colours at base fill also receive: `border-top: 1px solid rgba(255,255,255, 0.5)` and `box-shadow: 0 1px 2px rgba(0,0,0, 0.08)` to lift them slightly off the surface.

State Colours

State	Treatment
Active/selected section pill	<code>border: 1px solid #273038</code> , no fill change
Currently playing candidate card	Full border: <code>1px solid #273038</code> , PLAYING badge in <code>#273038</code> fill / <code>#FBFAF7</code> text
Generating progress dots	Three 4×4px squares: filled <code>#273038</code> → <code>#9A9B96</code> → <code>#CDCCC7</code> (left to right)
Locked item icon	<code>fa-lock</code> at 9px, <code>#273038</code>
Connected status dot	7px square , colour = agent’s semantic pastel (Reaper = <code>#C3B8D4</code>)

3. Typography

Typefaces

Role	Family	Source
Body / UI	DM Sans	Google Fonts — weights 300, 400, 500
Monospace / Data	IBM Plex Mono	Google Fonts — weights 400, 500

```
<link href="https://fonts.googleapis.com/css2?family=DM+Sans:wght@300;400;500&family=IBM+Plex+Mono:wght@400;500&display=swap" rel="stylesheet">
```

Type Scale

Token	Size	Weight	Tracking	Usage
type-hero	52px	300 (light)	-0.035em	New Session headline
type-h1	38px	300 (light)	-0.025em	Page-level headings (Audition)
type-h2	20px	400 (normal)	default	Panel headings
type-body-lg	17px	300	default	Intention text, form inputs
type-body	14–15px	300–400	default	Body copy, card descriptions
type-body-sm	13px	300–400	default	Rail labels, secondary copy
type-ui	11px	500	0.14–0.18em	Navigation links, button labels (uppercase)
type-label	10px	500	0.12–0.16em	Section labels (uppercase small caps style)
type-mono-lg	36px	500	default	A/B/C candidate label
type-mono-data	10px	400	0.16–0.18em	MIDI values, BPM, bar numbers, agent status
type-mono-xs	9px	400	0.10–0.12em	Grid annotations, timestamps, captions

Rules

- **DM Sans** carries all prose, navigation, UI labels, headings, and button text.
- **IBM Plex Mono** carries all MIDI data (note names, chord symbols, BPM, bar counts), technical status strings (VERSE · 5–8, COMMITTED, GENERATING), and all uppercase annotation tags.
- Uppercase labels use `letter-spacing: 0.12–0.18em`. Never use `font-variant: small-caps` — track manually.

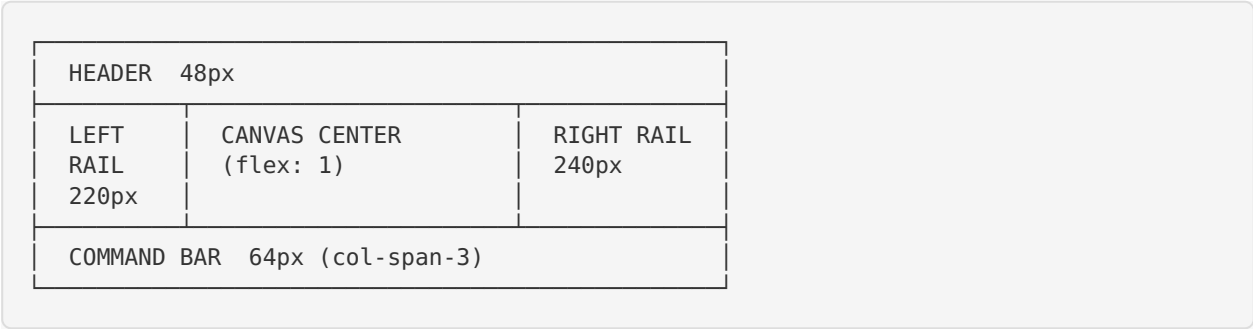
- Font weight 300 (light) is the default body weight. Weight 400 for card headings, navigation, active states. Weight 500 sparingly for UI labels and data emphasis.

4. Spacing & Grid

Base Unit

8px — all spacing, sizing, and layout measurements are multiples of 8.

Layout Grid (Desktop 1440px)



Zone	Width	Notes
Page	1440px fixed	Desktop-first, no responsive breakpoints in V0
Header	full-width × 48px	Fixed top, contains logo, song meta, nav, bridge status
Left Rail	220px	Session Agents, Locks
Right Rail	240px	Context, Selection, Intention, Lock state, Last Decision
Center Canvas	remaining (~980px)	MIDI piano-roll grid, section strip, playhead
Command Bar	full-width × 64px	Transport controls, natural-language input, Generate button

Common Spacing Values

Token	Value	Usage
space-1	4px	Micro gaps (dot spacing, inline gaps)
space-2	8px	Base unit — tight internal padding
space-3	12px	Card internal padding, icon-to-text gap
space-4	16px	Standard component padding
space-5	20px	Panel padding
space-6	24px	Section gap
space-8	32px	MIDI bar width on piano-roll
space-10	40px	Large panel padding
space-16	64px	Page section gap
space-24	96px	Page layout horizontal padding

5. Skeuomorphic Layer (The Core of Variation D)

This is a 40% skeuomorphic layer. All depth cues simulate a warm overhead light source at the top-left. No element exceeds 12px blur or 10% opacity on drop shadows (except the dark primary button).

Panel / Card Surfaces

Applied to all `bg-panel` (`#FBFAF7`) containers:

```
background: linear-gradient(160deg, #FFFFFF 0%, #F7F5F0 100%);
box-shadow:
  inset 0 1px 0 rgba(255, 255, 255, 0.6), /* top inner highlight */
  inset 1px 0 0 rgba(255, 255, 255, 0.3), /* left inner highlight */
  inset 0 -1px 0 rgba(0, 0, 0, 0.06), /* bottom inner shadow */
  inset -1px 0 0 rgba(0, 0, 0, 0.04), /* right inner shadow */
  0 1px 3px rgba(0, 0, 0, 0.06), /* near drop shadow */
  0 4px 12px rgba(0, 0, 0, 0.04); /* ambient lift */
```


Header Bar

```
box-shadow:
  0 1px 0 rgba(255, 255, 255, 0.85), /* bottom edge highlight */
  0 2px 6px rgba(0, 0, 0, 0.04); /* soft separation from content */
```

Dividers / Separators

Two-line engraved groove technique — do not use a single 1px flat line:

```
border-top: 1px solid rgba(0, 0, 0, 0.08);
box-shadow: 0 -1px 0 rgba(255, 255, 255, 0.85);
```

This reads as a groove pressed into the surface.

Default Button (Secondary)

```
background: linear-gradient(180deg, #F0EDE8 0%, #E0DDD8 100%);
box-shadow:
  inset 0 1px 0 rgba(255, 255, 255, 0.8), /* top highlight – lit face */
  0 2px 6px rgba(0, 0, 0, 0.10); /* drop shadow – lifted off surface */
transition: box-shadow 0.15s, background 0.15s;
```

Hover:

```
background: linear-gradient(180deg, #EDEAE4 0%, #E3E0DA 100%);
box-shadow:
  inset 0 1px 0 rgba(255, 255, 255, 0.6),
  0 3px 9px rgba(0, 0, 0, 0.10);
```

Primary / Commit Button (Dark)

The one high-contrast element. One per page.

```
background: linear-gradient(180deg, #1A1A1A 0%, #0D0D0D 100%);
color: #FBFAF7;
box-shadow:
  inset 0 1px 0 rgba(255, 255, 255, 0.08), /* subtle top glint */
  0 2px 8px rgba(0, 0, 0, 0.25);
```

Hover:

```
background: linear-gradient(180deg, #242424 0%, #111111 100%);
```

“Keep This” / Committed Button (Sage)

```
background: linear-gradient(180deg, #C3DDCD 0%, #ACCAB7 100%);
box-shadow:
  inset 0 1px 0 rgba(255, 255, 255, 0.55),
  0 2px 6px rgba(0, 0, 0, 0.10);
```

Input Fields & Textareas

Recessed / pressed-into-surface:

```
background: #EDEAE4;
box-shadow:
  inset 0 1px 3px rgba(0, 0, 0, 0.10),
  inset 0 1px 1px rgba(0, 0, 0, 0.06);
border: 1px solid #BEBB5;
```

Focus:

```
box-shadow: inset 0 1px 4px rgba(0, 0, 0, 0.14);
border-color: #273038;
```

MIDI Note Blocks

Each note block on the piano-roll (and candidate MIDI preview) uses the agent's semantic colour with micro-depth:

```
/* Example for Harmony note block */
background: linear-gradient(180deg, #BFCEDA 0%, #AEBECB 100%);
border-top: 1px solid rgba(255, 255, 255, 0.5); /* top highlight */
box-shadow: 0 1px 2px rgba(0, 0, 0, 0.08); /* lifts block off grid */
```

Apply same pattern to Bass (#DACFC0 → #C9BCAA), Drums (#DABFCA → #C9AAB6).

Agent Cards (Left Rail)

Each agent card receives a 3px left border in its semantic agent colour — acts as a physical card-slot tab:

```
/* Harmony */
border-left: 3px solid #B8C8D4;

/* Bass */
border-left: 3px solid #D4C8B8;

/* Drums */
border-left: 3px solid #D4B8C3;
```

Combined with the panel surface treatment, these read as tabbed card slots.

Section Blocks (Song Structure)

```
background: linear-gradient(180deg, [pastel-top] 0%, [pastel-bottom] 100%);
border: 1px solid rgba(39, 48, 56, 0.12);
box-shadow: 0 1px 4px rgba(0, 0, 0, 0.08); /* slightly raised tiles */
```

Page Background

```
background: radial-gradient(ellipse at 50% 0%, #F8F5F0 0%, #F0EDE6 100%);
```

Imperceptibly lighter at the top centre — implies overhead illumination across the whole canvas.

6. Component Library

Header Bar

- Height: 48px
- Background: #F5F2ED + header shadow treatment
- Left zone: 220px — logo mark (12×12px) + POPPIN wordmark in 11px / 500 / tracking-[0.16em]
- Center: song title (13px / 400) + monospace meta (BPM, KEY, METER at 10px)
- Right: navigation links (11px / uppercase / tracking-[0.12em]) + Reaper status dot + label
- Border-bottom: 1px solid #D8D4CE

Session Agent Card

```
[3px left accent border]
Agent name (13px/400)      ■ [dot]
STATUS (mono 9px)         8 BARS
```

- Background: panel surface treatment (#FBFAF7 + gradient + inset shadows)
- Border: 1px solid #D8D4CE
- Left accent: 3px solid [agent-colour]
- Status values: COMMITTED , GENERATING , IDLE
- Generating animation: 3× 4px squares in progression (#273038 → #9A9B96 → #CDCCC7)

MIDI Piano-Roll Canvas

- Grid background:


```
css
background-image:
  linear-gradient(to right, rgba(39,48,56,0.055) 1px, transparent 1px),
  linear-gradient(to bottom, rgba(39,48,56,0.05) 1px, transparent 1px);
background-size: 32px 100%, 100% 16px;
```
- Bar lines: 1px solid rgba(39,48,56,0.18) at every 32px horizontal interval
- Bar number labels: mono 9px / #9A9B97 flush left above track row
- Track label column: 88px wide, border-right: 1px solid #D8D4CE , labels in mono 9px
- Note blocks: height 12px , variable width by duration, agent colour with micro-gradient and top-highlight
- Playhead: 1px solid #273038 , 7×7px filled square at top

Command Bar

- Height: 64px , spans full width in 3-column grid
- Border-top: engraved divider treatment
- Left cell (220px): transport controls — Play / Loop / Stop as 32×32px bordered square buttons
- Center cell: monospace arrow → + text input (full-width, flat, 14px) + keyboard shortcut hint ⌘↵ + **Generate** button

- Right cell (240px): active agent status + generating dots
- Generate button: primary dark treatment (#273038 with dark gradient)

A/B/C Candidate Cards

- Layout: 3-column equal grid
- Each card: min-height ~514px , flexbox column
- Header zone: 80px — candidate letter (mono 36px / 500) + archetype name + density dots
- MIDI preview zone: 192px — mini piano-roll with agent-colour note blocks
- Description zone: flex-1 — WHY IT MOVES label (mono 9px) + explanation (14px) + note path (mono 9px)
- Action zone: border-top + 2-column button grid (Play/Loop + Keep This) + full-width Splice row
- **Playing state:** full border: 1px solid #273038 + PLAYING badge (#273038 fill, #FBFAF7 text, positioned -top-px right-5)
- **Inactive state:** border: 1px solid #BEBBB5

Density Dots

Three 8×8px squares indicating musical density (sparse → full):

● ○ ○ sparse
● ● ○ medium
● ● ● full

Filled: bg-[agent-colour] . Outlined: border: 1px solid [agent-colour] .

Commit History Row

[timestamp mono 9px] [AGENT badge] [section · bars] [description 13px] [↔ undo]

- Engraved border-top divider between rows
- Agent badge: 6px square in agent colour + agent name mono 9px
- Undo: 10px / tracking-[0.12em] uppercase / text-[#747A7D]
- Row hover: very subtle background shift to #F5F2ED

Section Strip (Composition Canvas top bar)

- Height: 56px , border-bottom: 1px solid #D8D4CE
- Section pills: height 28px , border: 1px solid #D8D4CE , mono 9px uppercase
- Active section: border: 1px solid #273038
- Format: SECTION-NAME · BAR-COUNT (e.g. VERSE · 8)

Reaper Status Indicator

- 7×7px square, colour = #C3B8D4 (lavender) when connected
- mono 10px uppercase label: REAPER / CONNECTED or REAPER / AUDITION READY
- Position: right zone of header

7. Motion & Transitions

Variation D uses restrained motion. Nothing animates without purpose.

Interaction	Duration	Easing	Property
Button hover	150ms	ease	background , box-shadow
Input focus	100ms	ease	box-shadow , border-color
Candidate card selection	150ms	ease	border-color , box-shadow
Agent status change	200ms	ease	opacity on status text
Generating dots	600ms loop	ease-in-out	opacity stagger (dot 1 → 2 → 3)
Playhead advance	real-time	linear	left position

No page transitions in V0. No entrance animations. No skeleton loaders.

8. Iconography

All icons: **Font Awesome 6.5.1 Solid** (fa-solid).

Icon	Usage	Size
<code>fa-play</code>	Transport play, candidate Play button	9px
<code>fa-pause</code>	Active candidate “Playing” state	8px
<code>fa-stop</code>	Transport stop	9px
<code>fa-repeat</code>	Transport loop	9px
<code>fa-rotate</code>	Regenerate all	8px
<code>fa-arrow-right</code>	Form submit, CTA	9px
<code>fa-arrow-left</code>	Back navigation	9px
<code>fa-chevron-down</code>	Dropdown indicator, Splice expander	8–9px
<code>fa-lock</code>	Locked material indicator	9px
<code>fa-square</code>	Agent status dot, density fill dot	7px

Agent status squares: `fa-solid fa-square` at 7px in the agent’s semantic colour. Idle states use `text-[#A3A29E]` .

Logo mark: not an icon — rendered as four `<span>` elements in a 2×2 CSS grid (see §1).

9. Page Inventory

Page	File	Primary Purpose
New Session	d-new-session.html	Start a session: BPM, key, meter, intention, reference
Composition Canvas	d-composition-canvas.html	MIDI piano-roll workspace — the central screen
A/B/C Candidate Audition	d-candidate-audition.html	Hear and choose between 3 generated takes
Song Structure	d-song-structure.html	Arrangement overview — all sections as a musical map
Commit History	d-commit-history.html	Chronological log of all decisions + undo
Reaper Bridge & Settings	d-reaper-settings.html	DAW connection status, sync, adapter config

Page Anatomy (All Pages)

Header (48px)

- └ Logo mark + wordmark | Song meta + nav | Reaper status

Content Area

- └ Left Rail (220px) – agents / locks / version tag
- └ Center Canvas (flex) – primary workspace
- └ Right Rail (240px) – context / selection / intention

Command Bar (64px, Composition Canvas only)

- └ Transport | Natural-language input + Generate | Agent status

10. Implementation Notes

Font Loading

```
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link href="https://fonts.googleapis.com/css2?family=DM+Sans:wght@300;400;500&family=IBM+Plex+Mono:wght@400;500&display=swap" rel="stylesheet">
```

CSS Custom Properties (Recommended)

```
:root {
  /* Backgrounds */
  --bg-page: #F5F2ED;
  --bg-panel: #FBFAF7;
  --bg-input: #EDEAE4;

  /* Text */
  --text-primary: #273038;
  --text-secondary: #646A6B;
  --text-tertiary: #747A7D;
  --text-muted: #8D908F;
  --text-faint: #999A96;
  --text-inverse: #FBFAF7;

  /* Borders */
  --border-default: #D8D4CE;
  --border-strong: #BEBBB5;
  --border-active: #273038;

  /* Agents */
  --colour-harmony: #B8C8D4;
  --colour-bass: #D4C8B8;
  --colour-drums: #D4B8C3;
  --colour-committed: #B8D4C3;
  --colour-reaper: #C3B8D4;

  /* Spacing */
  --space-base: 8px;

  /* Type */
  --font-body: 'DM Sans', sans-serif;
  --font-mono: 'IBM Plex Mono', monospace;
}
```

Skeuomorphic Layer Application Priority

1. Apply `body` radial gradient first (page background)
2. Apply panel gradient + inset + drop shadow to all `bg-panel` surfaces
3. Apply button gradient + shadow to all `<button>` elements; override with dark gradient for primary/commit buttons
4. Apply inset shadow + `bg-input` to all `<input>` and `<textarea>` elements
5. Apply engraved divider shadow to all `border-top` / `border-bottom` separator elements
6. Apply agent micro-gradient + top-highlight to all MIDI note blocks and agent-coloured surfaces
7. Apply agent left-accent border to all agent cards

Absolute Non-Negotiables

- No border-radius beyond what a specific component already uses (0px by default throughout).
- No drop shadow `blur` above 12px . No drop shadow `opacity` above 25% (except dark primary button at 25%).
- No new decorative elements. No illustrations. No stock photos.
- The primary commit action (Create Session, Generate, Keep This, Resync) is always the only high-contrast element on a given page.

- Monospace type is reserved exclusively for MIDI data, technical status strings, bar numbers, and timestamps. Never use it for prose.
- The outlined logo square is always transparent/open. Never fill it.

Poppin Music — You provide taste. The system provides technique.